

Of cars and toys

The single-molecule car built by Rice University in 2005 can be considered one of the first nanorobot successes, which paved the way for other molecular machines.

The nanocar consisted of a chassis and axles made of well-defined organic groups with pivoting suspension and freely rotating axles. The wheels were 'buckyballs' or spheres of pure carbon containing 60 atoms a piece. The entire car measured just 3-4 nanometres across, making it slightly wider than a strand of DNA.

Today, there are micrometre-sized toys on sale, such as the Hexbug Nano series from robot toymaker Innovation First.

Nanofibre-supported stem cell treatment has been experimentally proved by a few research teams across the world. One team of scientists has demonstrated its application in repairing heart cells, to help heart attack victims recover without requiring a heart transplant. Another research group has shown that nanofibre scaffolds can be used to overcome retinal degeneration. They implanted a disk-like scaffold into the eye, to guide the stem cells on where to grow the new retinal cells. This can be an effective cure for macular degeneration.

While most researchers are still working with scaffolds at the micro-level, it is believed that very soon we will have scaffolds that can work at the nano-level, making it possible to guide the stem cells more precisely on how to interact with existing cells. The technique could then be used to regenerate damaged spinal cords, cartilages in joints (for relief of arthritis) and nerve cells in the brain (to treat Parkinson's disease).

Experts rightly believe nanorobots to be the 'medicine of the future.' These tiny robots are made of components with micro- or nano-metre dimensions, generally 0.1-10 micrometres or smaller. They are often constructed with nanoscale or molecular components, and are capable of interacting with minuscule cells and other nanoscale structures. In one approach referred to as molecular machines, proteins and DNA could act as motors, mechanical joints, transmission elements or sensors. If all these different components are assembled together in proper proportion and orientation, they would form nano-devices with multiple degrees of freedom, able to

apply forces and manipulate objects in the nanoscale world.

The ability of nanorobots to interact with the building blocks of our body is likely to transform the face of disease detection and drug delivery, not to forget more effective surgical techniques. In the future, it is believed that powerful drugs will be combined with nanorobots and injected into the body or maybe prescribed even as a capsule, to identify affected areas and release the medicine effectively.

Flying into space too

Although medicine and healthcare appears to be the biggest market for nanorobots, these tiny bots are hopefully going to be very helpful in other fields as well. NASA, for instance, has been playing around for a long time with autonomous nanotechnology swarms (ANTS).

ANTS are basically groups of na-

norobots made with nano electromechanical systems and nano-sensors. In making these, the NASA team used nanotubes not only to make the robots smaller but also more flexible. Because struts made of metal tape and nanotubes are retractable, the robot can shrink until all its nodes touch.

With a spot of artificial intelligence, these swarms of nanorobots can move around and work together not just with fellow-ANTS but also with other swarms. The system can learn about and adapt to its environment, helping it to survive not just on earth but other planets too. The whole architecture is self-similar in that elements and sub-elements of the system may also be recursively structured as ANTS on scales ranging from microscopic to interplanetary distances.

For adding an element of intelligence to these swarms, researchers at the NASA Goddard Space Flight Centre are developing a software construct called neural basis function. The neural basis function will bridge the divide between lower- and higher-level functions and create bi-level intelligence capable of truly autonomous behaviour. It will greatly simplify the process of developing autonomous remote systems. A lower-level neural system would handle

basic system functions, security and safety, while a higher-level neural system would take care of problem solving, planning and scheduling. These two systems would interact via a third system called the evolvable neural interface that allows the artificial intelligence system to be situated in a real-world context.

These groups of bots can fall into formation, change their formation, move over un-



Autonomous nanotechnology swarms (ANTS)
(Image courtesy: <http://ants.gsfc.nasa.gov>)